

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

SAS821S – SECURITY ANALYTICS

SEMESTER 2 – 2026

MILESTONE 1

PROJECT CHARTER

**T06 – WATER UTILITY REMOTE ACCESS AND
OPERATIONAL-TECHNOLOGY ANOMALY
DETECTION**

Member A: Napandulwe Udeni Kafidi (212073273)

Role: Data and Modelling Lead

Member B: Hishikushitya Hipetwa (219048347)

Role: Security Engineering and Intelligence Lead

Submission Date: 10 August 2026

Assessment Weight: 20%

Contents

1. Group and Project Identification

Field	Group Response
Topic ID and approved topic	T06 – Water Utility Remote Access and Operational-Technology Anomaly Detection
Working project title	Detecting Suspicious Remote Access and OT Activity in a Water Utility
Member A	Napandulwe Udeni Kafidi (212073273) – Data and Modelling Lead
Member B	Hishikushitya Hipetwa (219048347) – Security Engineering and Intel-ligence Lead
Repository/location	Private GitHub repository and shared project folder – to be created after approval
Date submitted	10 August 2026

2. Problem and Decision Statement

Water utilities use operational technology (OT) to monitor and control pumps, valves, reservoir levels, water pressure and treatment processes. These systems must remain available because a disruption can affect water delivery, public health and the daily work of the utility. Internal technicians and external vendors may connect remotely to carry out maintenance. Remote access is useful, but it also creates a security risk. A stolen vendor password, an unmanaged device or an incorrectly configured connection could allow an unauthorised person to reach important OT assets.

The main problem is that normal maintenance activity and malicious activity can sometimes look similar. For example, a vendor may have a valid account, but a login at an unusual time, repeated failed attempts, a new source address or communication with several controllers may indicate that the account has been compromised. Changes in controller commands or sensor values shortly after such a login may provide further evidence. If these records are reviewed separately, the security team may miss the relationship between the events or respond too late.

The affected stakeholders include the security analyst, OT engineers, water operations staff, vendor managers and senior management. They need a clear view of the activity and a simple explanation of why an event was flagged.

The proposed project will create a small security analytics solution using synthetic vendor VPN logs, OT firewall records, sensor telemetry and maintenance-ticket text. Descriptive analysis will establish normal behaviour, while supervised and anomaly-detection methods will identify suspicious events. Related records will be joined into an incident timeline and displayed on a dashboard.

The main decision supported by the prototype is whether a remote session should be **allowed, monitored, investigated or blocked**. Where a high-risk event affects an important OT asset, the dashboard will also recommend actions such as verifying the vendor, ending the session, isolating the device and increasing monitoring. The project will use only controlled synthetic data and will not connect to or test a live water utility system.

3. Scope and Exclusions

In Scope	Out of Scope
Synthetic vendor VPN sessions, OT firewall events, controller/sensor telemetry and maintenance tickets; vendor accounts and selected OT assets; off-hours access, repeated failures, new devices or addresses, unusual protocols, abnormal commands and sensor changes; data cleaning, baselines, supervised and anomaly detection, text mining, incident timeline, control simulation, risk scoring and a Streamlit dashboard.	Live SCADA or water-utility access; penetration testing; control of physical pumps or valves; real customer, employee or vendor data; malware execution; full production deployment; advanced deep-learning models; replacement of operational or engineering judgement.

4. Objectives and Security Analytics Questions

Objective / Question	Measure of Completion
Objective 1	Generate and document four synthetic data sources containing at least 5,700 combined records.
Objective 2	Clean and join the data, produce a normal-behaviour baseline and create at least two useful security visualisations.
Objective 3	Train one supervised model and one anomaly method, then report a confusion matrix, precision, recall, F1-score and an interpretable anomaly threshold.
Objective 4	Create an evidence-based incident timeline, classify ticket text, compare three control scenarios and test three evasion or drift cases.
Objective 5	Build a working Streamlit dashboard that ranks alerts and recommends allow, monitor, investigate or block/isolate actions.
Question 1	What are the normal time, user, device, address, traffic and sensor patterns in the synthetic environment? Answered by descriptive statistics and baseline charts.
Question 2	Which remote sessions and OT events are suspicious? Answered by model predictions, anomaly scores and rule-based indicators.
Question 3	How are suspicious remote-access events connected to affected OT assets? Answered by a cross-source incident timeline and entity links.
Question 4	Which of the three proposed controls reduces risk most effectively? Answered by repeatable simulation results and a comparison table.

5. Learning Sessions 1–10 Coverage Matrix

Session Course Area	Planned Project Evidence
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1	Foundations of security analytics	Frame the remote-access problem and document the descriptive, diagnostic, predictive and prescriptive stages of the analytics lifecycle.
2	Analytical software and tools	Use Python, Jupyter, Pandas and Scikit-learn for reproducible ingestion, cleaning, transformation and analysis; use Streamlit for the prototype.
3	Machine-learning fundamentals	Design features, split labelled data into training and testing sets, and evaluate Random Forest and Isolation Forest outputs.
4	Machine learning in cybersecurity	Implement a practical detector that classifies suspicious vendor sessions and flags abnormal OT activity.
5	Analytics and incident response	Correlate VPN, firewall, telemetry and ticket evidence into a timeline, then recommend containment, recovery and monitoring actions.
6	Security intelligence	Define priority intelligence requirements and produce a short analyst alert plus an executive risk summary using enriched event context.
7	Access analytics	Build normal user, device, location and time baselines; detect off-hours access, new devices and unusual asset access.
8	Simulations and security processes	Run at least 1,000 iterations for each of three scenarios: current controls; MFA plus time restrictions; MFA plus segmentation and automatic isolation.
9	Security and text mining	Preprocess at least 250 maintenance tickets and use TF-IDF with Logistic Regression to classify normal maintenance, equipment faults and security concerns.
10	Advanced security intelligence	Produce a forward-looking session/asset risk score and test working-hours evasion, trusted-address misuse and slow low-volume activity or data drift.

6. Proposed Data Sources

Source	Format / Volume	Key Fields	Provenance / Security Controls
Vendor VPN authentication and session logs	CSV; 2,000 labelled records	session_id, time, user_id, vendor_id, source_ip, device_id, login_result, MFA, duration, bytes, label	Generated in Python from documented rules; pseudonymous IDs, non-routable addresses, no credentials or real people.
OT firewall and network logs	CSV; 2,000 unlabelled records	event_id, time, source, destination, asset_id, protocol, port, action, bytes, connection_count	Generated in Python to represent normal and unusual OT traffic; fictional assets and addresses; no live network data.
Controller and sensor telemetry	CSV; 1,500 semi-labelled records	time, asset_id, sensor_type, value, set_point, command_count, state, anomaly_flag	Generated ranges and injected anomalies documented in the data-generation script; no physical equipment connection.
Maintenance and security tickets	CSV; 250 unstructured text records	ticket_id, time, vendor_id, asset_id, description, category, priority	Synthetic ticket text created from documented templates; no names or personal details; reviewed for bias and unrealistic wording.

Total planned records: 5,750.

7. Proposed Analytical Methods

Component	Proposed Method / Tool	Expected Output
Baseline / EDA	Pandas, Matplotlib and Seaborn	Normal access baseline, failed-login and off-hours trends, traffic and sensor charts, plus descriptive statistics.
Supervised ML	Random Forest in Scikit-learn	Normal/suspicious prediction with confusion matrix, precision, recall and F1-score.
Unsupervised / UBA	Isolation Forest and simple behaviour rules	Anomaly score, documented threshold and ranked unusual users, devices, sessions and assets.
Incident investigation	Pandas time-window correlation	Incident timeline with event IDs, affected user/vendor, device and OT asset, followed by response actions.
Security intelligence	Priority intelligence requirements and event enrichment	Operational alert for analysts and a short executive risk summary with context and recommended action.
Simulation	Python Monte Carlo / what-if simulation	Comparison of three control scenarios using at least 1,000 iterations per scenario and stated assumptions.
Text mining / NLP	TF-IDF and Logistic Regression	Maintenance-ticket categories, important terms and classification evaluation.
Predictive / adversarial analysis	Weighted session/asset risk score and controlled manipulation tests	Forward-looking risk rating and measured effect of three evasion or drift cases on detection.
Prototype dashboard	Streamlit	Filterable alert list, risk cards, baseline charts, incident timeline and response recommendation.

8. High-Level Architecture

The solution will follow one simple path from synthetic data collection to an analyst decision. Python will clean and join the records using common IDs and timestamps. The prepared dataset will feed the detection, investigation, text-mining, simulation and risk modules. Their outputs will be combined in one Streamlit dashboard so that an analyst can view the evidence and choose an action.

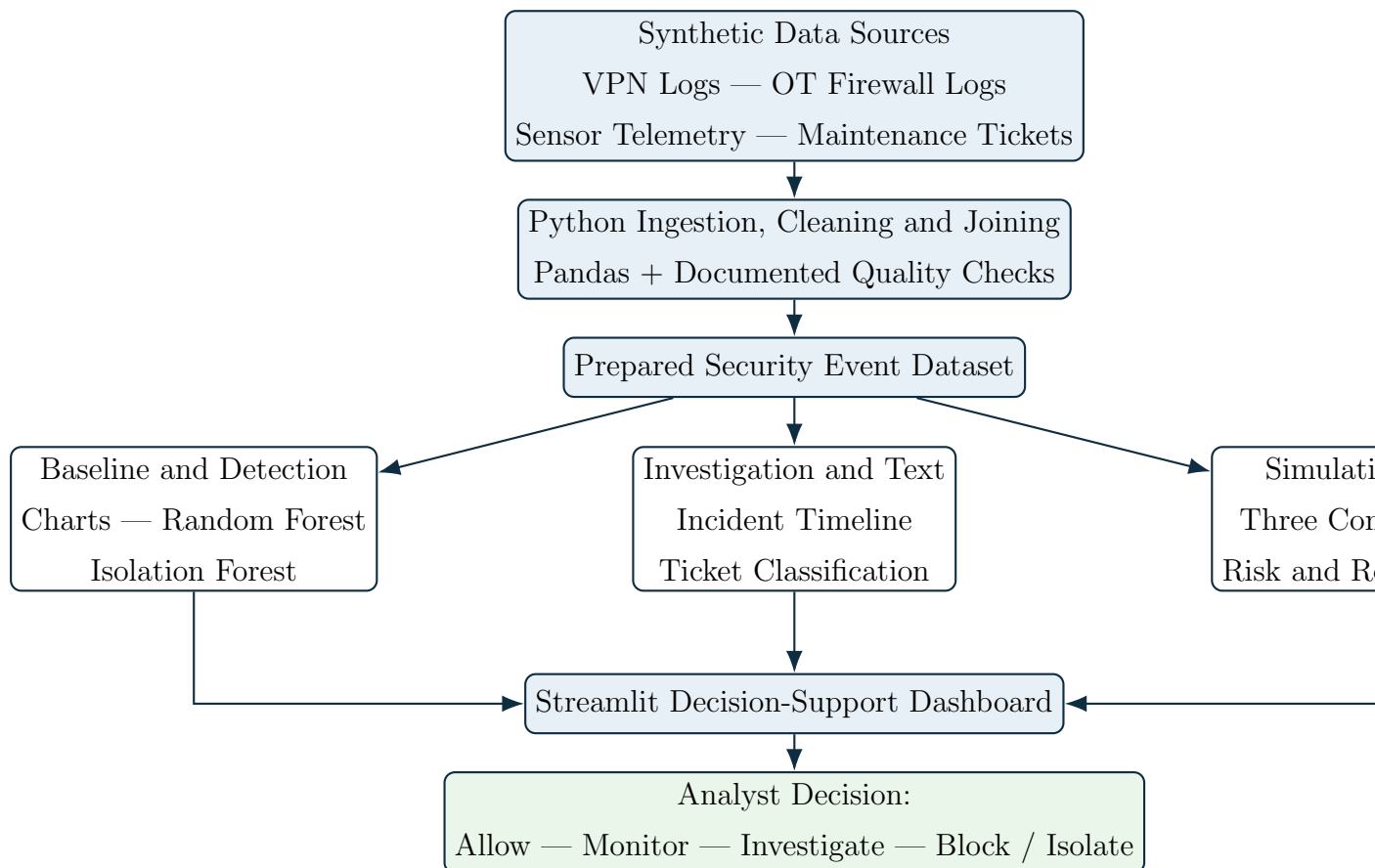


Figure 1: High-level security analytics decision-support architecture

9. Roles and RACI

R = Responsible, A = Accountable, C = Consulted, I = Informed

Workstream	Member A	Member B	Review / Approval Evidence
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Data acquisition and quality	A/R	C	Data dictionary, generation script and joint quality review
ML and anomaly modelling	A/R	C – security validation	Experiment log, metrics and partner review notes
Investigation and access analytics	C	A/R	Incident timeline, access findings and joint sign-off
Simulation and intelligence	R – analytical support	A/R	Scenario results, intelligence products and joint interpretation
Prototype and testing	R	R	Test cases, issue log and joint prototype approval
Documentation and presentation	R	R	Draft history, contribution log and presentation rehearsal

10. Work Plan

Task	Owner	Start	Finish	Dependency	Output
Finalise and submit project charter	Both	9 Aug	10 Aug	Topic allocation	Submitted charter and agreed data design
Generate, clean and document data	Member A	11 Aug	23 Aug	Submitted charter	Four datasets, data dictionary and cleaning log
Baseline, ML and anomaly modelling	A with B review	24 Aug	5 Sep	Prepared data	Charts, model metrics and anomaly list

Investigation, intelligence simulation	NLP, and support	B with A	6 Sep	12 Sep	Model outputs	out-	Timeline, results, intelligence and scenario comparison	ticket intel-briefs
Integrate dashboard and test prototype		Both	13 Sep	19 Sep	All analytical modules		Working Streamlit dashboard, test log and repository snapshot	
Submit implementation, final report and presentation		Both	20 Sep	30 Sep	Tested prototype	proto-	Milestones and contribution evidence	2–4

11. Risk and Ethics Register

Risk	Likelihood	Impact	Mitigation	Owner
Data unavailable or unsuitable	Medium	High	Use documented synthetic generation rules, check class balance and request facilitator approval early.	A
Privacy or sensitive-data exposure	Low	High	Use only fictional users, assets and addresses; keep the repository private and remove credentials or secrets.	Both
Model bias or misleading results	Medium	Medium	Review class balance, use several metrics, inspect false alerts and explain that results are based on synthetic data.	A
Scope or time overrun	High	High	Use simple models, complete the required modules first, keep a weekly task board and move optional features to future work.	Both

Technical integra- tion failure	Medium	Medium	Build separate modules with com- mon fields, test early and keep a recorded demonstration as a fall- back.	B
Unequal contribu- tion	Medium	High	Use assigned roles, shared reviews, meeting notes, issue assignments and version-control commits.	Both

12. Responsible AI and Academic Integrity Statement

The group will use only authorised synthetic data, public references and properly licensed software. Generative AI may be used, where permitted, for brainstorming, explaining concepts, language editing and coding guidance. It will not be treated as evidence. Both members will review, test and understand all code, calculations, figures and written claims before submission. Public datasets, external code and references will be cited, and any reused code will be checked for licence and security risks. The group will keep a short record of permitted AI assistance and disclose it in the final appendix. No generated reference, result or claim will be included unless it has been verified by the group.

13. Group Agreement and Declaration

- We confirm that this group consists of exactly two students.
- We accept the role allocation and will retain evidence of our contributions.
- We understand that both members must explain the complete solution during the presentation and Q&A.
- We will use only authorised data and will not test live third-party systems.
- We will disclose permitted AI assistance, reused code and external data sources.

Member	Signature	Date
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Member A – Napan- dulwe Udeni Kafidi	_____	10 August 2026
Member B – Hishikushitya Hipetwa	_____	10 August 2026

Facilitator approval is requested for the use of fully synthetic datasets and the proposed tools and methods.

14. Lecturer Approval

Decision	Comments / Conditions	Lecturer / Date

15. References

1. Namibia University of Science and Technology. (2026). *SAS821S Security Analytics: Capstone Project 2026*.